

Construction of a rotation

To construct a rotation, three elements of information are needed: the object to be rotated, the location of the center of rotation, and the size of the angle of rotation.

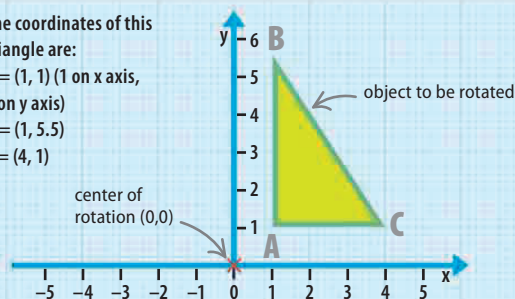
The coordinates of this triangle are:

$A = (1, 1)$ (1 on x axis,

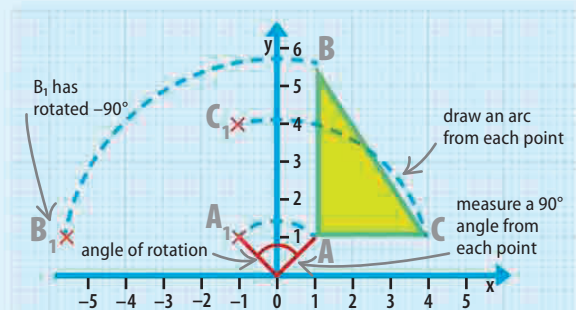
1 on y axis)

$B = (1, 5.5)$

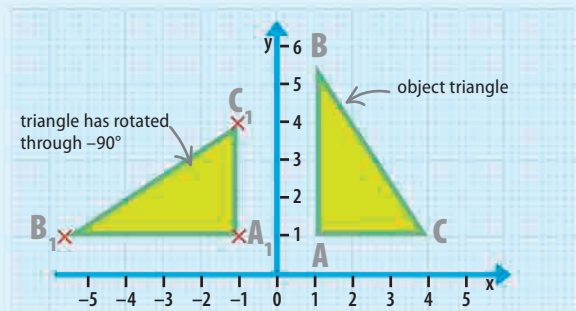
$C = (4, 1)$



Given the position of the triangle ABC (see above) and the center of rotation, rotate the triangle -90° , which means 90° counterclockwise. The image of triangle ABC will be on the left-hand side of the y axis.



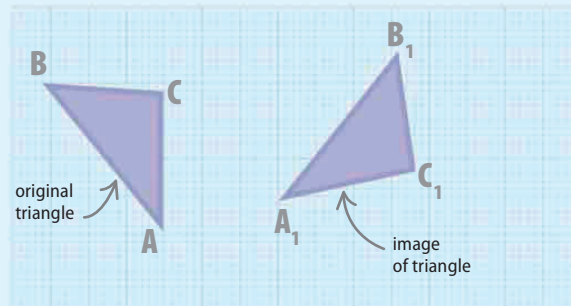
Place a compass point on the center of rotation and draw arcs counterclockwise from points A , B , and C (counterclockwise because the rotation is negative). Then, placing the center of a protractor over the center of rotation, measure an angle of 90° from each point. Mark the point where the angle meets the arc.



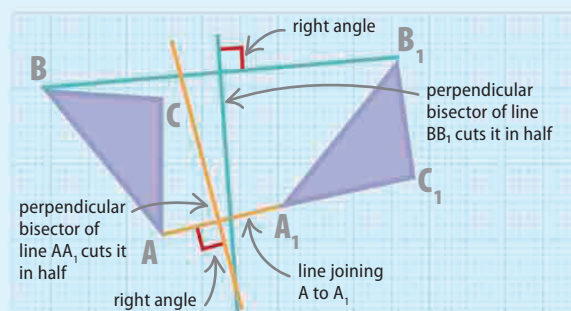
Label the new points A_1 , B_1 , and C_1 . Join them to form the image. Each point on the new triangle $A_1B_1C_1$ has rotated 90° counterclockwise from each point on the original triangle ABC .

Finding the angle and center of a rotation

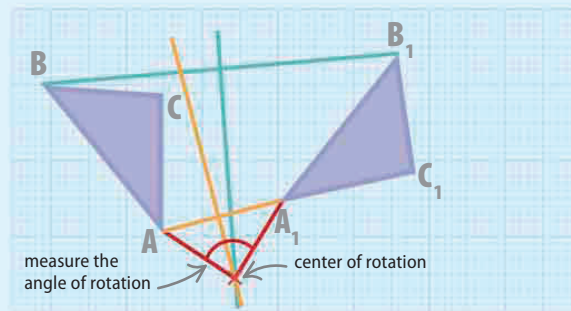
Given an object and its rotated image, the center and angle of rotation can be found.



The triangle $A_1B_1C_1$ is the image of triangle ABC after a rotation. The center and angle of rotation can be found by drawing the perpendicular bisectors (lines that cut exactly through the middle—see pp.110–111) of the lines between two sets of points, here A and A_1 and B and B_1 .



Using a compass and a ruler, construct the perpendicular bisector of the line joining A and A_1 and the perpendicular bisector of the line that joins B and B_1 . These bisectors will cross each other.



The center of rotation is the point where the two perpendicular bisectors cross. To find the angle of rotation, join A and A_1 to the center of rotation and measure the angle between these lines.